The Pre-FOMC Drift Lost Its Morning

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Question

Lucca and Moench (2015) found that US stocks earned a large excess return in the 24 hours before scheduled FOMC announcements, measured from 2pm on the previous day to 2pm on the announcement day. Studies using daily closing prices report that the drift weakened or disappeared after 2015. Is the drift gone, and if not, when during the 24 hours is it now earned?

Result

The pre-announcement return has not disappeared. Its timing has changed. Over September 2003 to March 2011, most of it accrued during the trading morning of the announcement day, between the open and 2pm. After March 2011 that morning component is zero. The overnight component, from the previous close to the announcement-day open, has stayed at about 13 to 17 basis points and is now essentially the whole drift. Closing-price studies understate it, because the hours after the announcement have averaged weak returns since 2011. Figure 1 shows the cumulative abnormal return path in the two periods.

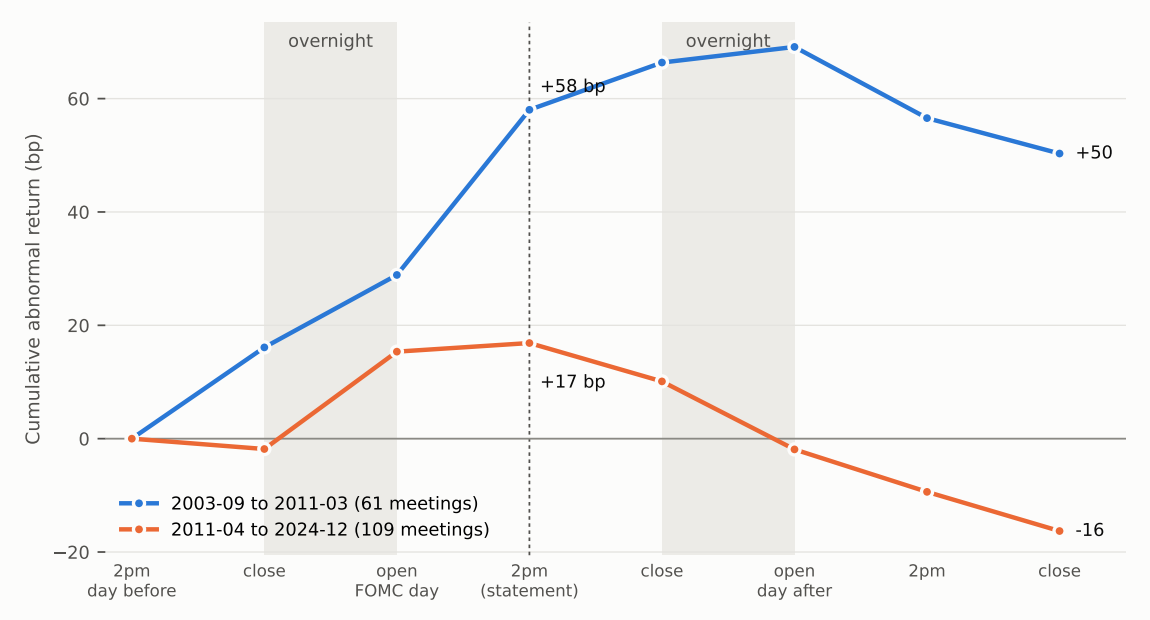


Figure 1: Cumulative abnormal SPY return from 2pm on the day before a scheduled FOMC announcement to the close of the day after, by period. Shaded areas are overnight intervals; the dotted line marks 2pm on the announcement day.

Data and design

Sample. SPY, the S&P 500 exchange-traded fund, from September 2003 to December 2024: 5,319 trading days, including 170 scheduled FOMC announcement days (61 from September 2003 to March 2011, 109 from April 2011 to December 2024). Announcement dates are the final days of scheduled meetings listed on the Federal Reserve Board's website; unscheduled meetings and conference calls are excluded.

Prices. Open and close are CRSP daily prices. The 2pm price is the mid-point of the national best bid and offer in the last quote at or before 13:59:59, from TAQ. Every 2pm quote on an announcement day is time-stamped 13:59:48 or later, so it precedes the statement (2:15pm before 2013 and 2:00pm afterwards).

Legs. For each day t, the return from 2pm on day $t-1$ to the close on day $t+1$ is split into seven price returns: the afternoon of $t-1$ (2pm to close), the night into t (close to open), the morning of t (open to 2pm), the afternoon of t (2pm to close), the night into $t+1$, the morning of $t+1$, and the afternoon of $t+1$.

Estimator. For each leg, the abnormal return is the mean on announcement days minus the mean on all other days, in basis points. Standard errors allow unequal variances across the two groups. The change between periods is the interaction coefficient in a regression of the leg return on an announcement dummy, a post-March-2011 dummy and their product, with heteroskedasticity-robust standard errors.

	2003-09 to 2011-03		2011-04 to 2024-12		Change	
Leg	bp	t	bp	t	bp	t
Day before, 2pm to close	16.1	1.0	−1.8	−0.5	−17.9	−1.1
Night into announcement day	12.8	1.5	17.2	3.7	4.4	0.4
Announcement day, open to 2pm	29.1	4.3	1.5	0.4	−27.6	−3.6
2pm-to-2pm window (sum of the above)	58.0	2.9	16.9	2.6	−41.2	−2.0
Announcement day, 2pm to close	8.3	0.7	−6.8	−0.7	−15.1	−1.0
Night into the next day	2.8	0.3	−12.0	−1.6	−14.8	−1.2
Next day, open to 2pm	−12.6	−1.1	−7.5	−1.1	5.1	0.4
Next day, 2pm to close	−6.3	−1.0	−6.9	−1.5	−0.7	−0.1
Announcement days	61		109			

Table 1: Abnormal SPY price returns around scheduled FOMC announcements: mean on announcement days minus the mean of the same leg on all other days, in basis points. Period columns: t-statistics with unequal-variance standard errors. Change column: interaction coefficient with heteroskedasticity-robust standard errors.

The morning leg carried most of the drift in the first period (29 bp, $t = 4.3$) and vanished in the second (1.5 bp); the fall of 28 bp has $t = -3.6$. The overnight leg is similar in both periods and is significant in the second (17 bp, $t = 3.7$). It holds separately in April 2011 to December 2015 (19 bp, $t = 2.4$) and January 2016 to December 2024 (16 bp, $t = 2.8$). Since April 2011, the roughly 3% of nights that precede an announcement account for 21% of SPY's cumulative overnight return.

Checks

- **Price source.** The overnight result is the same with TAQ official auction prices (16.8 bp, $t = 3.5$) and with quote mid-points just before the close and just after the open (16.4 bp, $t = 3.5$), and after adding dividends on ex-dates (no announcement day is an SPY ex-dividend date).
- **Outliers.** Excluding 2020 gives 13.7 bp ($t = 3.1$); winsorizing at the 1st and 99th percentiles gives 16.4 bp ($t = 3.6$). The median overnight return before an announcement is 9.7 bp, and 68% are positive against 54% on other nights. Several of the largest values coincide with other news, such as 5 November 2020 during the US election count.
- **Placebos.** Nights before Wednesdays without an announcement show 1.1 bp ($t = 0.4$); the night two days before an announcement shows 3.9 bp ($t = 0.6$).
- **Announcement timing.** In 2011 and 2012 the statement at press-conference meetings was released at 12:30pm, inside the morning leg. Dropping those eight meetings leaves the morning leg at −0.6 bp and the overnight leg at 15.4 bp ($t = 3.2$).

Caveats

The sample starts in September 2003, so it covers only the second half of Lucca and Moench's sample. Returns are SPY price returns rather than S&P 500 futures returns. The overnight leg can include

macroeconomic releases at 8:30am on announcement days; the one large overlap identified (the CPI release on 12 June 2024) does not change the result, but a complete release calendar was not used. Seven legs were examined in each period, so individual weak results should not be read as findings. The post-announcement legs are negative in the second period but none is individually significant.

Relation to the literature

Kurov, Wolfe and Gilbert (2021) use daily data and report that the drift weakened after 2015. Boguth, Grégoire and Martineau (2019) relate the post-2011 drift to press conferences. Pan and Peng (2024) report that the equity drift occurs mostly overnight and that a Treasury drift precedes it. This note adds the split by period: the morning component disappeared after 2011 while the overnight component did not. We have not established whether this split is reported elsewhere.

References

[1] Boguth, O., Grégoire, V., and Martineau, C. (2019). Shaping Expectations and Coordinating Attention: The Unintended Consequences of FOMC Press Conferences. *Journal of Financial and Quantitative Analysis* 54(6), 2327–2353.

[2] Kurov, A., Wolfe, M. H., and Gilbert, T. (2021). The Disappearing Pre-FOMC Announcement Drift. *Finance Research Letters* 40.

[3] Lucca, D. O., and Moench, E. (2015). The Pre-FOMC Announcement Drift. *Journal of Finance* 70(1), 329–371.

[4] Pan, J., and Peng, Q. (2024). The Pre-FOMC Drift in Long-Term Treasury Bonds. Working paper, SSRN 4764451.

Data: CRSP daily stock file and TAQ quotes via WRDS; FOMC dates from the Federal Reserve Board. Only aggregate statistics are reported. Drafted with AI assistance and checked before publication.